Andrea Perissinotto

Senior C++ Software Engineer | Unreal Engine & Unity

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Experience

ZURU TECH ITALY | Senior Software Developer

September 2023 -- Present | Modena, IT

- Develop robust C++ features for a real-time 3D application built with Unreal Engine, working extensively with Actors, Components, and UObjects.
- Design geometry-driven algorithms and sustainable workflows, supporting art and technical features while collaborating with multidisciplinary teams.
- Optimize performance and maintain clean architectural standards across complex spatial and interactive systems.

VECTION ITALY SRL | Senior Software Developer

March 2021 -- August 2023 | Casalecchio di Reno, IT

- Developed real-time 3D applications for industrial use cases using Unity and C#, targeting AR (HoloLens 2) and VR (Oculus/HTC Vive).
- Coordinated 3D asset integration and collaboration between software developers and 3D artists.
- Managed technical development activities, task estimation, and delivery using Agile methodologies.

TEL-AVIV UNIVERSITY | Senior Software Developer

March 2019 -- March 2021 | Tel-Aviv, IL

- Led the development of CellStudio, a C++ agent-based simulator for complex immunological systems.
- Collaborated with CERN openlab on parallelization infrastructure for computationally intensive simulations.
- Designed simulation architecture, translated scientific requirements into software solutions, and mentored students.

GLANCE VISION TECHNOLOGIES (GVT) SRL | Software Developer

April 2016 -- January 2019 | Trieste, IT

- Led the development of RoboSim, a 3D industrial robotics simulator integrating C++, C#, and Unity.
- Designed and implemented algorithms for industrial applications and real-time 3D simulation.

INTERNATIONAL SCHOOL FOR ADVANCED STUDIES (SISSA) | Algorithm Developer

August 2015 -- April 2018 | Trieste, IT

- Designed and implemented scientific computing, image processing, and data analysis algorithms in collaboration with researchers.

Selected Projects

ROBOSIM | C++, C#, Unity

2018

- Developed a real-time 3D robotic simulation integrating a C# Windows Forms application with a Unity runtime through bidirectional communication.
- Implemented robot trajectory editing, interactive scene control, and realistic varnishing-effect simulation.

CELLSTUDIO | C++

2019--2020

- Developed a large-scale agent-based simulation using configurable state machines and multithreaded execution.
- Integrated 3D visualization through Cinder.

Technical Skills

PROGRAMMING

C++ • C#
Python • Matlab
JavaScript • LaTeX

3D DEVELOPMENT

Unreal Engine • Unity • OpenCV

C++ & ENGINEERING

Templates • Smart Pointers
Multithreading • Algorithms
Data-Oriented Design•
Unity ECS/Burst Compiler •
State Machines (FSM) •
Game Loops•
Simulation •
Computational Geometry •
Software Architecture

TOOLS

Git • Gitlab • Trello

Education

UNIVERSITÀ DI BOLOGNA

MSc Biomedical Engineering

2010 -- 2013 | Cesena, IT

Grade: 106/110

UNIVERSITÀ DI TRIESTE

BSc Electronic Engineering

2006 -- 2010 | Trieste, IT

Grade: 102/110

Research

ICVS

Multimodal image registration

2014 -- 2015 | Braga, PT

ZHAW

SCATh -- Smart Catheterization

2010 -- 2013 | Winterthur, CH